

Life Cycle Impact Assessment — External Load-Bearing Wall

Scope A1–C3 · Functional unit: 1 m² · Target U-value ≤ 0.20 W/m²K ·
ÖKOBAUDAT OBD_2024_I_A2

Note on terminology: The question refers to "life cycle cost" but requests GWP, PERT, and PENRT — these are environmental impact indicators, not monetary costs. This answer interprets the task as a *life cycle environmental impact assessment* per EN 15804+A2.

SECTION 01

Results Summary

FUNCTIONAL UNIT: 1 M ² EXTERNAL LOAD-BEARING WALL · SCOPE: A1–C3	
GWP total	39.38 kg CO ₂ -eq / m ²
PENRT (non-renewable primary energy)	456.25 MJ / m ²
PERT (renewable primary energy)	105.05 MJ / m ²
U-value (verified)	0.196 W/m ² K ≈ 0.20

Scope: **A1** **A2** **A3** **C3** included where declared per EPD. C4 and Module D excluded. C3 was not declared for mineral wool and lime-cement plaster in the selected datasets; therefore, it was treated as zero for this simplified estimate.

SECTION 02

Assumed Wall Build-up

#	LAYER (INSIDE → OUTSIDE)	GERMAN DATASET NAME	THICKNESS	DECLARED UNIT
1	Interior plaster	GIPSPUTZ/Tool	15 mm	1 kg
2	Load-bearing masonry	Mauerziegel (ungefüllt)	240 mm	1 m³
3	Thermal insulation	Mineralwolle-Dämmstoff im niedrigen Rohdichtebereich	160 mm	1 m³
4	Exterior render	Lime-cement plaster / Kalkzementputz	20 mm	1 m³
Total wall thickness			435 mm	

435 mm is within the typical range for German external load-bearing masonry with external insulation (300–500 mm).

SECTION 03

ÖKOBAUDAT Datasets Used

MATERIAL	SUBTYPE	DENSITY	VALID UNTIL	UUID / EPD-NO.
Mauerziegel (ungefüllt) 30514538-fcb4-483b-b5d5-c108d2037536	average	575 kg/m³	2026	EPD-BDZ-20210066
GIPSPUTZ/Tool 5618a4e7-1f60-4205-9869-30c2e9fad094	average	1050 kg/m³	2027	EPD-BVG-20210317
Mineralwolle-Dämmstoff (niedriger Rohdichtebereich) f721d148-f727-4a55-b2e6-52ecb90e09be	average	24 kg/m³	2028	EPD-FMI-20210016
Sphera/BBSR				

MATERIAL	SUBTYPE	DENSITY	VALID UNTIL	UUID / EPD-NO.
Lime-cement plaster / Kalkzementputz dca9f05c-036b-40e5-ae81-c957c2420fd8	generic	1800 kg/m³	2026	

SECTION 04

LCA Calculation per 1 m² Wall — Scope A1-C3

LAYER	QTY / M²	GWP A1-A3	GWP C3	GWP TOTAL
Gypsum plaster	15.75 kg	2.27	0.00	2.27
Brick (Mauerziegel)	0.240 m³	27.12	-2.42	24.70
Mineral wool	0.160 m³	5.52	0.00 †	5.52
Lime-cement plaster	0.020 m³	6.90	0.00 †	6.90
TOTAL GWP (A1-C3)		41.81	-2.42	39.38 kg CO₂-eq

LAYER	PENRT A1-A3	PENRT C3	PENRT TOTAL	PERT A1-A3	PERT C3	PERT TOTAL
Gypsum plaster	33.67	0.00	33.67	2.19	0.00	2.19
Brick	283.20	6.50	289.70	51.60	0.55	52.15
Mineral wool	89.38	0.00 †	89.38	19.57	0.00 †	19.57
	43.50	0.00 †	43.50	31.14	0.00 †	31.14

LAYER	PENRT A1–A3	PENRT C3	PENRT TOTAL	PERT A1– A3	PERT C3	PERT TOTAL
Lime- cement plaster						
TOTAL	449.75	6.50	456.25	104.50	0.55	105.05
(A1–C3)			MJ			MJ

† C3 was not declared for mineral wool and lime-cement plaster in the selected datasets; therefore, it was treated as zero for this simplified estimate.

Brick C3 GWP = $-10.1 \text{ kg CO}_2\text{-eq/m}^3$ per EPD scenario S1 (94% recycling of demolished brick).

SECTION 05

Assumptions

- Functional unit: 1 m² external load-bearing masonry wall, 50-year service life
- Scope: A1–A3 (production) + C3 (waste processing) where declared. C4 and Module D excluded.
- C3 was not declared for mineral wool or lime-cement plaster in the selected datasets; therefore, it was treated as zero for this simplified estimate.
- Brick: 240 mm, standard load-bearing thickness per DIN 1053 / Eurocode 6
- Mineral wool: 160 mm, $\lambda = 0.035 \text{ W/mK}$ — thickness set to achieve $U \leq 0.20 \text{ W/m}^2\text{K}$
- Gypsum plaster density 1050 kg/m³ per EPD declared value → 15.75 kg per m² at 15 mm
- All datasets from ÖKOBAUDAT OBD_2024_I_A2 — generic/average datasets preferred
- No windows, openings, mortar joints, or fixings included — homogeneous wall panel only

- Transport stages (A4, C2) not included — outside question scope and insufficient project data

APPENDIX A

U-Value Verification — ISO 6946

Thermal transmittance calculated per **ISO 6946:2017**. Lambda (λ) values from **DIN 4108-4** and EPD declared values. Surface resistances per ISO 6946 Table 1 (horizontal heat flow, external wall).

LAYER	D (M)	λ (W/MK)	SOURCE	R = D/ λ (M ² K/W)
Rsi — internal surface resistance	—	—	ISO 6946 Table 1	0.130
Gypsum plaster	0.015	0.35	DIN 4108-4	0.043
Brick (Mauerziegel, 575 kg/m ³)	0.240	0.80	DIN 4108-4 Table 1	0.300
Mineral wool insulation	0.160	0.035	EPD declared value	4.571
Lime-cement plaster	0.020	0.87	EPD range 0.8-0.9, midpoint used	0.023
Rse — external surface resistance	—	—	ISO 6946 Table 1	0.040
Total R				5.107 m ² K/W

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// ISO 6946:2017 – Thermal transmittance of building components
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R_total = Rsi + R_gypsum + R_brick + R_wool + R_render + Rse
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R_total = 0.130 + 0.043 + 0.300 + 4.571 + 0.023 + 0.040
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R_total = 5.107 m2K/W
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$$U = 1 / R_{\text{total}} = 1 / 5.107$$

$$U = 0.196 \text{ W/m}^2\text{K} \checkmark \approx 0.20 \text{ W/m}^2\text{K}$$

Result satisfies the GEG 2020 (Gebäudeenergiegesetz) requirement of $U \leq 0.24 \text{ W/m}^2\text{K}$ for external walls and meets the target $U \leq 0.20 \text{ W/m}^2\text{K}$ specified in the question.